

Product Requirements Document (PRD) — PrepGenie

1. Product Vision

PrepGenie is an AI career-prep copilot that gives every student instant, judgment-free feedback on their resume and interview performance — the kind of mentorship most students can't otherwise access.

2. Problem Statement

Most B.Tech students:

- Don't get personalized resume reviews before applying to internships.
- Have no safe space to practice mock interviews before the real one.
- Don't know which skills/topics to prioritize based on their target role.

3. Target Users

Persona	Need
Pre-final year B.Tech student	Wants ATS-friendly resume before placement season
Final year student prepping for interviews	Wants realistic mock interview practice
Career switcher / non-CS student	Wants to know what skills/gaps to fix

4. Goals & Non-Goals

Goals

- Fast, accurate AI resume feedback (structure, ATS score, keyword gaps)
- Role-specific mock interview simulation with scoring
- Visual progress tracking over multiple sessions

Non-Goals (to avoid scope creep)

- No real video-based interview analysis (facial expression/emotion detection) — text/voice-to-text only
- No job-board integration or auto-apply features
- No payment/subscription system (v1 is free, single-tier)
- No multi-language support in v1 (English only)

5. Core Features (v1 Scope)

5.1 Authentication

- Email/password + Google OAuth via Supabase Auth
- Protected routes for dashboard, analyzer, interview

5.2 Resume Analyzer

- Upload PDF/DOCX resume
- Extract raw text (pdf-parse/mammoth)
- Send to Claude API with structured prompt → returns:
 - ATS Compatibility Score (0-100)
 - Missing keywords for target role
 - Section-wise suggestions (Summary, Skills, Projects, Experience)
 - Grammar/clarity issues
- Store analysis result in DB linked to user

5.3 Mock Interview Coach

- User selects: Role (SDE Intern / Data Analyst / Core Engineering / HR Round)
- AI generates 5-8 questions dynamically based on role + resume content
- One question shown at a time (chat-style UI)
- User answers via text (voice-to-text optional stretch goal)
- AI scores each answer: Correctness, Clarity, Confidence (via structured rubric), plus one-line improvement tip
- End-of-session summary score + weak topics

5.4 Dashboard

- Resume score history (line chart)
- Interview performance by category (radar/bar chart)
- Weak topics list (auto-aggregated from feedback)
- Streak tracker (days active)

5.5 Settings/Profile

- Update target role
- View/download past resume analyses
- Delete account/data (privacy compliance)

6. Functional Requirements

ID	Requirement
FR1	System must parse PDF/DOCX and extract clean text within 5s
FR2	System must call AI API with resume text + return structured JSON response
FR3	System must persist all resume analyses per user with timestamp
FR4	System must generate interview questions dynamically (not hardcoded)
FR5	System must score answers using a consistent rubric (JSON schema enforced)
FR6	Dashboard must visualize at least 2 data types via charts
FR7	All AI calls must have loading states + error fallback UI

7. Non-Functional Requirements

- Response time: AI feedback within 8-10 seconds (show loading skeletons)
- Security: No API keys exposed client-side; all AI calls proxied via backend
- Reliability: Handle malformed resumes / empty uploads gracefully
- Accessibility: Keyboard navigable, proper contrast ratios
- Responsiveness: Mobile, tablet, desktop layouts

8. Data Model (Simplified)

```
users (id, email, name, target_role, created_at)
resumes (id, user_id, file_name, raw_text, ats_score, feedback_json,
created_at)
interview_sessions (id, user_id, role, questions_json, answers_json,
scores_json, created_at)
progress_snapshots (id, user_id, resume_score, interview_score,
snapshot_date)
```

9. AI Prompting Strategy

- Use strict system prompts enforcing **JSON-only output** for both resume analysis and interview scoring so the frontend can render structured UI reliably.
- Use few-shot examples in the system prompt for consistent scoring rubrics.
- Temperature kept low (0.3-0.4) for scoring consistency; slightly higher (0.6-0.7) for question generation variety.

10. Risks & Mitigations

Risk	Mitigation
AI returns malformed JSON	Add JSON schema validation + retry logic
Resume parsing fails on scanned/image PDFs	Show clear error: "Please upload a text-based PDF"
API rate limits during demo	Cache last analysis result client-side
Scope creep during build	Strictly follow day-by-day blueprint; defer stretch features to "Future Scope"

11. Success Criteria for v1 Launch

- End-to-end flow (signup → upload resume → get analysis → do mock interview → see dashboard) works with zero critical bugs
- Deployed and publicly accessible
- README makes the project understandable in under 2 minutes to a recruiter